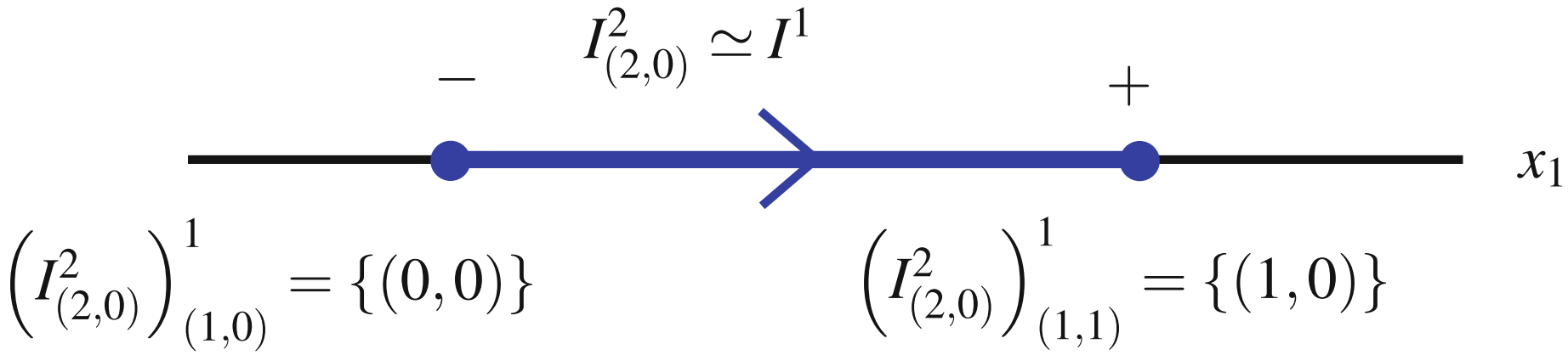

$I^2_{(2,0)} \simeq I^1$
$-$
$+$
x_1
$\left(I^2_{(2,0)}\right)^1_{(1,0)} = \{(0,0)\}$
$\left(I^2_{(2,0)}\right)^1_{(1,1)} = \{(1,0)\}$